

Enhanced durability assessment of GFRP and HFRP bars embedded in mortar under simulated seawater exposure

Yunfeng Pan^a, Junmin Wang^a, Angelo Paulo Mogul^b, Jun He^{c,*}, Gaojie Xu^a

^a School of Civil Engineering and Architecture, Zhejiang Sci-Tech University, Hangzhou, 310018, China

^b College of Architecture and Fine Arts, Bulacan State University, Malolos City, 3000, Philippines

^c Key Laboratory of Low-carbon Urban Construction of Xi'an, Chang'an University, Xi'an, 710061, China

ARTICLE INFO

Keywords:

Carbon/glass hybrid fiber reinforced polymer bar
Glass fiber reinforced polymer bar
Interlaminar shear strength
Saline-alkaline solution
Durability

ABSTRACT

Glass fiber in the GFRP bar is prone to erosion by the alkaline solution in concrete. To improve the durability of the GFRP bars in marine structures, a low-alkali MPC mortar and carbon fiber coating wrapping core GFRP layer was proposed. This study examined the effects of these protective measures on the durability of GFRP bars by evaluating interlaminar shear strength degradation through short beam shear tests. Both the low-alkali mortar and carbon fiber coating effectively enhanced durability under simulated seawater exposure. The interlaminar shear strength of GFRP bars embedded in OPC mortar decreased more severely than those embedded in MPC mortar. A carbon fiber coating wrapped around the GFRP bar significantly enhanced its durability when embedded in mortar in seawater. After 200 days exposure at 60 °C, GFRP bars in MPC exhibit 201.5 % higher strength retention compared to those in OPC, while carbon fiber coated HFRP bars retain 34.7 % more strength than GFRP bars. Additionally, GFRP and HFRP bars embedded in OPC mortar retained higher interlaminar shear strength compared to those directly immersed in a saline-alkaline solution. When compared to field-exposed GFRP bars at 4 °C, degradation predictions for bare GFRP bars immersed in alkali-salt solution tended to overestimate interlaminar shear strength loss. In contrast, predictions for GFRP bars embedded in OPC mortar slightly underestimated the degradation. According to the prediction, the estimated exposure time required to reach 70 % interlaminar shear strength retention at 4.0 °C is as follows: 5.4 years for bare GFRP bars, 97.5 years for bare HFRP bars, 59.0 years for GFRP bars in OPC mortar, and 487.6 years for HFRP bars in OPC mortar.

1. Introduction

Glass fiber reinforced polymer (GFRP) bar has been widely used as a replacement for steel bar in marine engineering structures, addressing the structural failure caused by steel corrosion due to chloride ions [1,2]. After 120 days of exposure under simulated marine conditions, the yield strength of steel decreased by 31 % [3], whereas the tensile strength of GFRP bars exhibited a reduction of only 9.0 % [4]. Typically, GFRP bars reinforce concrete structures, but the highly alkaline pore solution of ordinary Portland cement concrete (OPC) erodes the glass fibers, leading to a degradation of GFRP bars' macro-mechanical properties [5]. To improve the long-term durability of GFRP bar in concrete, two strategies have been proposed: reducing the alkalinity of the internal environment

* Corresponding author.

E-mail address: hejun@chd.edu.cn (J. He).

during service and preventing direct exposure of glass fibers to the concrete pore solution [6–8]. Consequently, the durability mechanisms of GFRP and HFRP bars embedded in concrete under marine conditions have become a significant research focus.

GFRP bars deteriorate in seawater due to the combined effects of salt ions, water molecules and OH^- from the concrete pore solution [5,9,10]. Their tensile strength retention is lower in alkaline environments than in saline solutions [11], with OH^- ions having a more pronounced impact than salt ions. In an alkaline solution ($\text{pH} = 12.6$), the tensile strength of 13-mm GFRP bars declined by 20.1 % after six months [12]. The degradation rate accelerates over time, as shown by a 1 % and 11 % reduction in interlaminar shear strength after 42 and 208 days of alkaline exposure at 60 °C, respectively [13]. Seawater exposure at 55 °C led to tensile strength retention rates of 87.3 %, 76.4 %, and 66.7 % after 3, 6, and 9 months, respectively [14]. Elevated temperatures exacerbate the detrimental effects of alkaline solutions on GFRP bars; for instance, immersion at 21 °C and 55 °C for 84 days resulted in shear strength reductions of 7 % and 68 %, respectively [15].

The degradation of the GFRP bar in saline-alkaline environments involves in glass fiber erosion, resin matrix hydrolysis, and fiber-matrix debonding. The deterioration begins at the surface and progresses inward [14]. Resin matrix hydrolysis weakens fiber bonding, exposing the glass fibers to direct OH^- ion attack, which erodes the silicon dioxide (-Si-O-Si-) structure. Environmental temperature fluctuations and hygrothermal stresses induce differential strain between the glass fibers and the resin matrix, leading to interface debonding [16,17]. Newly formed cracks further facilitate OH^- ion migration, accelerating the breakdown of fibers, the resin matrix, and their interfacial bond, ultimately reducing the mechanical properties of GFRP bars.

The degradation of the GFRP bar in alkaline solution intensifies with increasing alkalinity. After 140 days at 40 °C, GFRP bars immersed in solutions with pH values of 12.7, 13, and 13.4 exhibited interlaminar shear strength reductions of 30 %, 32 %, and 39 %, respectively [18]. After 12 months at 60 °C, tensile strength decreased by 45.9 % at pH 13, whereas at pH 11, the reduction was 27.6 % [7]. However, changes in alkalinity have minimal effects on the resin matrix itself; tensile strength reductions of resin samples immersed in solutions with pH levels of 7.5, 10, 12.6 after 180 days were 32.6 %, 31.9 %, and 35.7 %, respectively [19].

GFRP bars directly immersed in alkaline solutions experience more severe mechanical property degradation than those embedded in concrete [20]. In practical applications, GFRP bars are typically embedded in concrete, which provides some protection. Estimates suggest that 7.5 years of degradation in concrete at full moisture levels is equivalent to one year of exposure in an alkaline solution [11]. Compared to traditional aging in simulated pore solutions, mortar-wrapped GFRP bars exhibit significantly lower tensile strength loss and reduced microstructural/physical deterioration [21]. The effects of mortar wrapping warrant further investigation.

Encasing GFRP bars with a carbon fiber coating to form HFRP bars is an effective strategy for improving durability in harsh environments. While most research on carbon/glass fiber hybridization focuses on mechanical properties and deformation capacity [22], the high cost of carbon fiber limits widespread use in civil engineering. To address this, we proposed an alternative approach: enhancing GFRP durability by externally wrapping a thin carbon fiber layer. Compared to GFRP bars, HFRP bars retained 34 % and 15 % higher shear strength after 100 days in an alkaline solution at 40 °C and 60 °C, respectively [6]. The elastic modulus of GFRP and HFRP bars does not differ significantly, and its influence on the overall mechanical performance is limited, primarily due to the thin carbon fiber layer on the surface [23]. Jin et al. [24] reported that substituting part of the basalt fibers with carbon fibers in basalt/carbon HFRPs improved tensile strength retention by 15.7 % after 180 days in OPC mortar under seawater exposure. Research suggests that in harsher immersion environments, HFRP bars exhibit higher interlaminar shear strength retention. Under alkaline and salt-alkaline conditions at 60 °C, retention rates of HFRP bars were 42.8 % and 61.1 %, respectively [25].

This study investigates two approaches for enhancing the durability of GFRP bars in concrete: reducing the alkalinity of the internal service environment and preventing direct contact between glass fibers and the pore solution through protective carbon fiber coatings. Experimental analysis evaluated the effects of alkalinity and carbon fiber coatings on GFRP durability. Compared to previous studies on GFRP and HFRP bars directly immersed in saline-alkaline solutions, this research specifically examined the influence of mortar wrapping on durability degradation. Degradation mechanisms under different influencing factors were analyzed using SEM and MIP techniques. Based on the experimental results, long-term mechanical performance predictions for GFRP and HFRP bars were developed and validated against data from actual service environments.

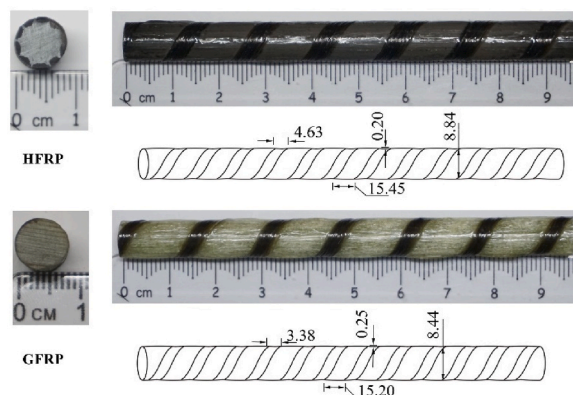


Fig. 1. Morphology of GFRP and HFRP bars(unit: mm).

2. Experimental

2.1. Raw materials

Fig. 1 shows GFRP and HFRP bars, both designed by Harbin Institute of Technology and manufactured by Harbin FRP Institute of China. The same epoxy resin, consisting of Bisphenol-A resin and Methylhexahydrophthalic Anhydride hardener, supplied by Bluestar New Chemical Materials Co., Ltd. (Wuxi, China), was used as the matrix material for both GFRP and HFRP bars. These FRP bars were prepared through a pultrusion process, which a forming temperature of 180 °C and a pulling speed of 40 cm/min. The nominal diameter of the FRP bars is 8 mm. The tensile strength of GFRP and HFRP bar are tested to be 1.1 GPa and 1.3 GPa, respectively. The specific parameters of fibers, resin matrix, and FRP bars were obtained from reference [6].

To study the effect of mortar with different pH values on the degradation performance of FRP bars, two types of mortar were selected: magnesium phosphate cement mortar (MPC) and ordinary Portland cement mortar (OPC). The mix ratio for MPC mortar is as follows: water: magnesium oxide: ammonium dihydrogen phosphate: silica fume: borax: sand = 1:5.27:1.67:1:0.40:9.33. The mix ratio for OPC is water: cement: sand = 1:2.25:2.95, with P.O 42.5 ordinary Portland cement used. The average particle size (d₅₀) of the magnesium oxide, ammonium dihydrogen phosphate and borax is 88.4 μm, 35.8 μm and 65.1 μm, respectively. The fineness modulus of the sand is 2.7.

2.2. Preparation of FRP bar specimens wrapped in mortar

The geometrical dimensions of FRP bar embedded in mortar are shown in Fig. 2a. A PVC mold with an outer diameter of 60 mm was used, with the FRP bar placed at the center. The bars were wrapped with either OPC mortar or MPC mortar. The FRP bar embedded in mortar had a length of 600 mm, with a mortar layer thickness of 26 mm [26]. Borax, ammonium dihydrogen phosphate, and sand are mixed with water and stirred for 2 min to ensure all materials are fully wetted. Subsequently, magnesium oxide and silica fume are gradually added according to the predetermined ratios, and the mixture is stirred slowly for another 2 min. The OPC mortar preparation involved a standard mixing protocol: 30 s of dry component homogenization preceding a 90-s wet mixing stage post-water addition. The freshly mixed mortar was poured into PVA pipes, which were removed 48 h after laboratory curing.

The FRP bar embedded in mortar is shown in Fig. 2b. Since the diffusion of water molecules along the fiber direction is much greater than that perpendicular to the fiber direction [27], and given that the length of the FRP bars is much greater than their diameter, the diffusion of water molecules is typically considered only perpendicular to the fiber direction. Therefore, epoxy adhesive was used to seal both ends of the FRP bar embedded in mortar, preventing the diffusion of water molecules along the fiber direction.

2.3. Exposure conditions

The specimens were subjected to a simulated environment of the internal conditions of concrete under marine service conditions. The simulated seawater was prepared according to the ASTM D1141-98 (2021) standard, and the reagent amounts used to prepare the seawater are shown in Table 1. Based on the aging conditions of FRP bars in the literature, three immersion temperatures were used at 21 °C, 40 °C, and 60 °C [28].

The immersion temperature was controlled using an HHI-600 electric constant-temperature water bath (Suzhou Donghua Testing Instruments Co., Ltd.), as shown in Fig. 3a. Polyethylene bags were used to isolate the water bath solution from the simulated seawater solution, as shown in Fig. 3b. The immersion durations were 30, 70, 120, and 200 days. After reaching the specified age, the specimens were removed, the outer layer of concrete was shattered, and FRP samples were cut into short beam shear specimens.

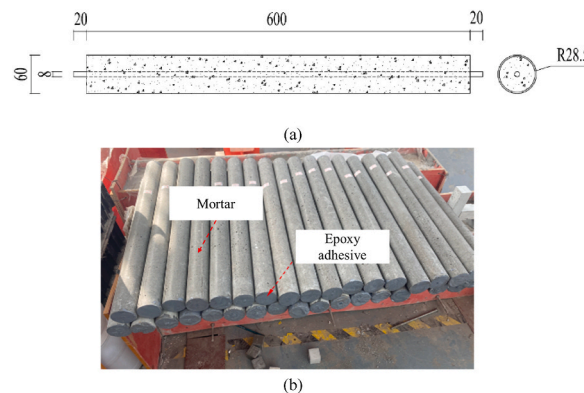


Fig. 2. (a) Geometrical dimensions and (b) photo of mortar-encapsulated FRP bar specimens.

Table 1
Reagent quantities for preparing simulated seawater (per 100L).

Reagent	Mass(g)
NaCl	2453.4
Na ₂ SO ₄	409.4
MgCl ₂ ·6H ₂ O	1111.2
CaCl ₂	115.8
SrCl ₂	4.2
KCl	69.5
NaHCO ₃	20.1
KBr	10.0
H ₃ BO ₃	2.7
NaF	0.3



Fig. 3. Exposure conditions, (a) electric constant-temperature water bath, (b) specimens in simulated seawater.

2.4. Short beam shear test

According to the recommended testing method in ASTM D4475-02(2016), the interlaminar shear strength was determined using the short beam shear tests, which characterize the Interfacial adhesion strength at the fiber-matrix interface. The test setup is shown in Fig. 4. A universal testing machine (CMT4304) was used, applying load under displacement control at a speed of 1.3 mm/min. An aspect ratio of 1/5 was used, with the support distance under the loading platform set to $5D = 40$ mm ($D = 8$ mm, representing the FRP bar diameter). Each group was repeated with four specimens.

The maximum interlaminar shear stress of FRP bars can be calculated using equation (1), as specified in ASTM D4475-02(2016).

$$\tau_{\max} = 0.849P/D^2 \quad (1)$$

where $\tau_{\max}$ represents the maximum interlaminar shear stress, also referred to as the interlaminar shear strength (MPa). P is the maximum applied load (kN).

The naming convention for the specimens in this study is as follows:

"G" and "H" represent GFRP and HFRP, respectively. The number following the FRP bar type indicates the immersion temperature. For example, 21, 40, and 60 represent environmental temperatures of 21 °C, 40 °C, and 60 °C, respectively. "P" and "M" represent OPC mortar and MPC mortar, respectively. The number following the mortar type indicates the number of immersion days. The immersion ages are as follows: 0 days (control group), 30 days, 70 days, 120 days, and 200 days.

For example, "H-21-P-30" indicates an HFRP bar embedded in OPC at an environmental temperature of 21 °C, aged for 30 days.

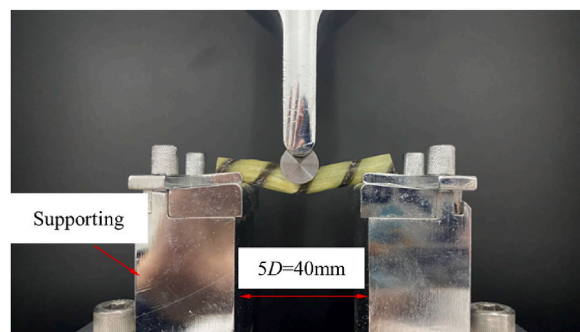


Fig. 4. Short beam shear test fixture.

2.5. Scanning electron microscope (SEM) analysis

A SEM (Zeiss Gemini SEM 500, Germany) was used to examine the resin and fiber distribution at the failure site of the FRP bar after the short beam shear test. This analysis aimed to explain the degradation mechanism of FRP bars and its impact on mechanical properties.

2.6. Mercury intrusion porosimetry (MIP) analysis

The effects of the seawater immersion on the porosity of the mortar were investigated using MIP with a MicroActive AutoPore V9600 model. The MIP analysis was performed using a step-wise pressure application approach and followed Berodier's research [29]. This method ensures controlled intrusion measurements while minimizing potential pore structure alterations. Each sample was tested only once due to experimental constraints.

3. Results and discussion

3.1. Failure mode

Fig. 5 presents the failure modes of GFRP and HFRP bars embedded in mortar before and after aging, following the short beam shear tests. Fig. 5a and b illustrate that the failure of unaged GFRP and HFRP bars occurred at the neutral axis, where a primary crack propagated along the fiber direction, ultimately leading to interlaminar shear failure.

Fig. 5c–j presents the failure modes of GFRP and HFRP bars embedded in OPC at different aging durations under 21 °C seawater immersion. It can be observed that under the 21 °C aging conditions, the failure modes of both GFRP and HFRP bars embedded in OPC

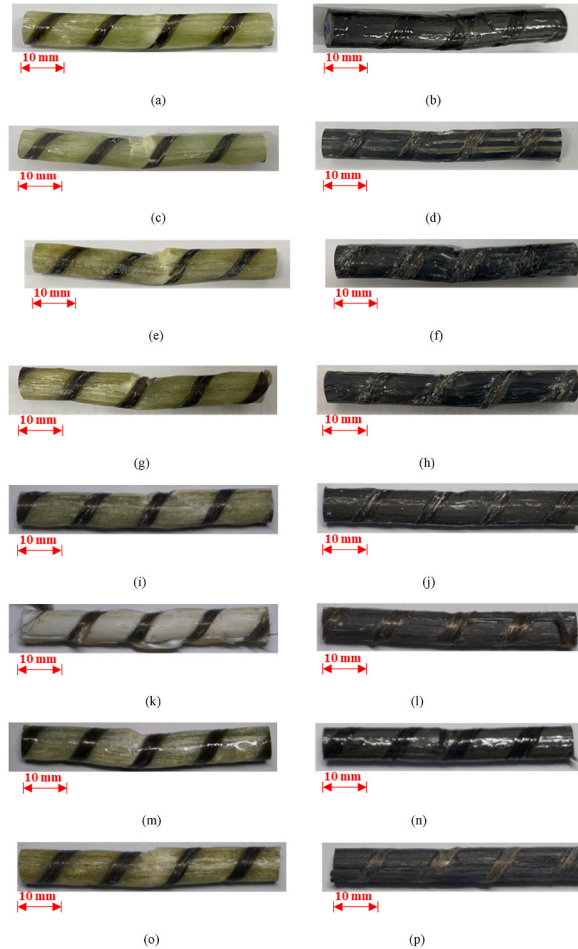


Fig. 5. Failure mode of GFRP bar and HFRPbar, (a) control GFRP bar, (b) control HFRP bar, (c) G-21-P-30, (d) H-21-P-30, (e) G-21-P-70, (f) H-21-P-70, (g) G-21-P-120, (h) H-21-P-120, (i) G-21-P-200, (j) H-21-P-200, (k) G-60-P-200, (l) H-60-P-200, (m) G-21-M-200, (n) H-21-M-200, (o) G-60-M-200 and (p) H-60-M-200.

are similar to those of the unaged bars, with failure occurring due to the propagation of shear cracks at the neutral axis of the FRP bars. This indicates that at 21 °C seawater immersion aging conditions, prolonged exposure does not affect the failure pattern of OPC-embedded GFRP and HFRP bars.

Fig. 5i–l shows the failure modes of GFRP and HFRP bars embedded in OPC at 200 days of aging under 21 °C and 60 °C seawater immersion. In Fig. 5i, GFRP bars exposed to 21 °C seawater for 200 days maintain an intact surface. In contrast, Fig. 5k depicts the failure mode of GFRP bars subjected to 60 °C seawater immersion for the same duration. It can be seen that a significant amount of resin has hydrolyzed on the surface, exposing the fibers. The surface has become loose, and numerous cracks have developed along the fiber direction.

A comparison of Fig. 5i and k reveals that higher temperatures accelerate the surface deterioration of GFRP bars. Fig. 5l shows the failure mode of HFRP bars in OPC after 200 days of seawater immersion at 60 °C. Some degree of resin degradation is evident, with the fibers exposed. However, compared to Fig. 5k and l, the surface deterioration of the HFRP bars is less severe. The elevated temperature at 60 °C increased the diffusion rate of water molecules and OH⁻ ions in the pore solution, hastening the mechanical degradation of GFRP bars. Despite this, the outer carbon fiber layer in HFRP bars helped slow down the degradation process.

Table 2

Test results of interlaminar shear strength of FRP bars.

Specimens	Type of mortar	Immersion duration (days)	Environmental temperature (°C)	Interlaminar shear strength (MPa)	Cv
G-21-P-0	OPC	0	—	55.3	2.7
G-21-P-30	OPC	30	21	48.8	1.3
G-21-P-70	OPC	70	21	57.4	3.9
G-21-P-120	OPC	120	21	52.6	0.3
G-21-P-200	OPC	200	21	57.8	2.8
G-40-P-30	OPC	30	40	51.1	5.0
G-40-P-70	OPC	70	40	54.3	2.0
G-40-P-120	OPC	120	40	46.9	1.9
G-40-P-200	OPC	200	40	45.8	1.5
G-60-P-30	OPC	30	60	46.9	1.9
G-60-P-70	OPC	70	60	41.8	1.4
G-60-P-120	OPC	120	60	29.0	0.5
G-60-P-200	OPC	200	60	19.6	1.9
G-21-M-0	MPC	0	—	55.3	2.7
G-21-M-30	MPC	30	21	53.2	1.1
G-21-M-70	MPC	70	21	57.0	2.0
G-21-M-120	MPC	120	21	52.2	4.7
G-21-M-200	MPC	200	21	54.5	1.9
G-40-M-30	MPC	30	40	51.8	1.6
G-40-M-70	MPC	70	40	55.3	5.7
G-40-M-120	MPC	120	40	55.4	1.7
G-40-M-200	MPC	200	40	53.7	1.7
G-60-M-30	MPC	30	60	53.9	2.2
G-60-M-70	MPC	70	60	54.0	3.8
G-60-M-120	MPC	120	60	54.2	3.7
G-60-M-200	MPC	200	60	59.1	5.0
H-21-P-0	OPC	0	—	54.3	2.1
H-21-P-30	OPC	30	21	63.4	1.6
H-21-P-70	OPC	70	21	61.8	1.9
H-21-P-120	OPC	120	21	60.8	0.0
H-21-P-200	OPC	200	21	59.7	1.6
H-40-P-30	OPC	30	40	61.3	2.6
H-40-P-70	OPC	70	40	56.9	6.6
H-40-P-120	OPC	120	40	59.2	0.5
H-40-P-200	OPC	200	40	56.2	2.3
H-60-P-30	OPC	30	60	60.4	4.1
H-60-P-70	OPC	70	60	65.1	3.9
H-60-P-120	OPC	120	60	46.8	1.6
H-60-P-200	OPC	200	60	38.2	4.1
H-21-M-0	MPC	0	—	54.7	0.4
H-21-M-30	MPC	30	21	61.4	3.8
H-21-M-70	MPC	70	21	63.5	3.2
H-21-M-120	MPC	120	21	54.2	4.5
H-21-M-200	MPC	200	21	57.8	1.1
H-40-M-30	MPC	30	40	57.7	2.5
H-40-M-70	MPC	70	40	56.1	3.1
H-40-M-120	MPC	120	40	55.6	0.6
H-40-M-200	MPC	200	40	57.0	0.8
H-60-M-30	MPC	30	60	60.6	0.4
H-60-M-70	MPC	70	60	65.0	0.6
H-60-M-120	MPC	120	60	56.6	0.5
H-60-M-200	MPC	200	60	49.7	6.0

Fig. 5i, j, m, and n depict the failure modes of GFRP and HFRP bars embedded in mortar at 200 days of aging under 21 °C seawater immersion. In comparison to the FRP bars embedded in OPC, the surface of the FRP bars embedded in MPC remains glossy and intact, with no visible signs of hydrolysis. Fig. 5k, l, o, and p show the failure modes of GFRP and HFRP bars embedded in mortar after interlaminar shear tests at 200 days of aging under 60 °C seawater immersion.

Regardless of the mortar type, some resin hydrolysis occurs on the surfaces of GFRP and HFRP bars. However, degradation is less severe in bars embedded in MPC compared to those in OPC. With a curing pH of approximately 9.0–9.6, MPC mortar creates a lower-alkali environment, reducing the degradation rate of FRP bars during service [30].

3.2. Effects of the alkalinity in mortar on the interlaminar shear strength

Table 2 presents the interlaminar shear strength of all specimens, while Fig. 6 illustrates the impact of mortar alkalinity on the shear strength of GFRP and HFRP bars. As shown in Fig. 6a, the shear strength of GFRP bars in OPC declines as environmental temperature rises. After 200 days, compared to unaged specimens, GFRP bars in OPC exhibit reductions of −4.5 %, 17.2 %, and 64.6 % at 21 °C, 40 °C, and 60 °C, respectively. Higher temperatures accelerate the reaction between OH^- ions in OPC and the resin matrix, as well as glass fibers, weakening the fiber-resin interface and leading to degradation [31]. The interlaminar shear strength of the GFRP bars embedded in OPC at 40 °C and 60 °C decreases with the immersion time, whereas immersion duration has little effect at 21 °C. As exposure continues, OH^- ions and water molecules penetrate from the surface, breaking down the resin, glass fibers, and interfacial bond, leading to degradation. At 21 °C, the slower diffusion of corrosive ions and water results in a weaker reaction with glass fibers and the resin matrix, causing minimal changes in shear strength.

Compared to GFRP bars in OPC, those embedded in MPC exhibit minimal variation in interlaminar shear strength, as shown in Fig. 6a. After 200 days, the shear strength of GFRP bars in MPC at 21 °C, 40 °C, and 60 °C decreased by 1.4 %, 2.9 %, and −6.9 %, respectively, relative to unaged samples. In contrast, compared to GFRP bars in OPC, those in MPC showed increases of −5.7 %, 17.2 %, and 201.5 % at the same temperatures. This difference is attributed to MPC's lower alkalinity, with a pH range of 9.0–9.6 compared to OPC's 12.5–13. Studies confirm that MPC helps reduce the internal pH of concrete surrounding GFRP bars, thereby enhancing their durability in seawater environments.

Fig. 6b presents the interlaminar shear strength of HFRP bars embedded in mortar. During the first 30 days of seawater immersion, a strength increase is observed, with HFRP bars in OPC and MPC gaining 11 %–17 % and 6 %–13 %, respectively, across all tested temperatures. Beyond 30 days, the impact of environmental temperature at 21 °C and 40 °C remains minimal, with OPC-embedded HFRP bars losing −9.1 % and −3.4 % of their strength after 200 days, while those in MPC decline by −6.1 % and −4.7 %, respectively. After 200 days, seawater immersion at 60 °C causes a more pronounced deterioration, with interlaminar shear strength reductions of 29.7 % in OPC and 8.5 % in MPC. These findings highlight the superior durability of HFRP bars wrapped with low-alkali MPC in mitigating degradation of the inner GFRP.

3.3. Effects of the carbon fiber coating on the interlaminar shear strength

Fig. 7 illustrates the impact of carbon fiber coating on FRP bars in mortar. As shown in Fig. 7a, for bars embedded in OPC, the interlaminar shear strength of GFRP bars deteriorates more significantly than that of HFRP bars. After 200 days at 21 °C, 40 °C, and 60 °C, the shear strength retention of HFRP bars is 3.3 %, 22.7 %, and 94.9 % higher, respectively, compared to GFRP bars, demonstrating the protective effect of the carbon fiber coating in highly alkaline environments. Fig. 7b presents the influence of carbon fiber coating on FRP bars embedded in MPC, where seawater immersion has a less pronounced effect on both GFRP and HFRP bars. After 200 days, the interlaminar shear strength of GFRP bars decreases by 1.4 %, 2.9 %, and −6.9 % at 21 °C, 40 °C, and 60 °C, respectively, while HFRP bars exhibits reductions of −6.4 %, −5.0 %, and 8.5 % under the same conditions. These findings highlight the superior protective role of low-alkali MPC mortar, minimizing mechanical strength degradation in both FRP types. This indicates that while the carbon fiber coating significantly improves the durability of GFRP bars, its effect on HFRP bars embedded in MPC

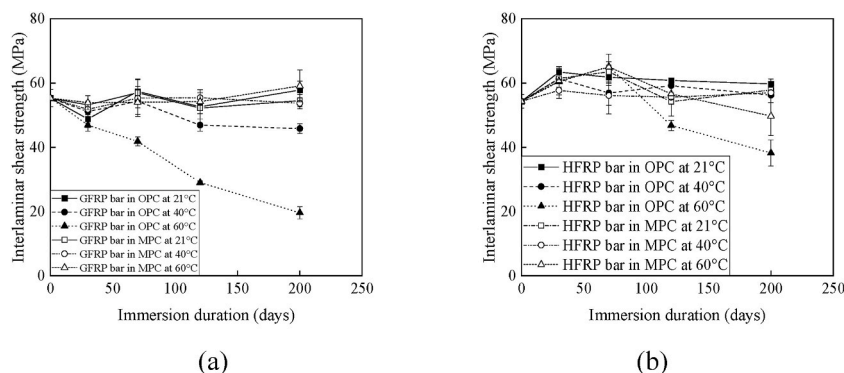


Fig. 6. Effects of the alkalinity in mortar on the interlaminar shear strength of (a) GFRP bar and (b) HFRP bar.

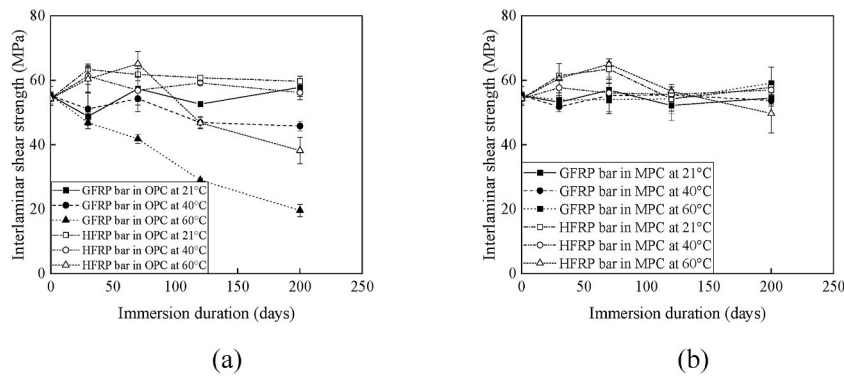


Fig. 7. Effects of the carbon fiber coating on the (a) OPC and (b) MPC-wrapped FRP bars.

remains minimal.

3.4. Effects of mortar wrapping on the interlaminar shear strength

Fig. 8 illustrates the impact of mortar wrapping on interlaminar shear strength. A bare FRP bar was immersed in a simulated alkaline solution replicating the pore environment of OPC mortar, as referenced in a previous study [25]. Fig. 8a presents the interlaminar shear strength retention of FRP bars at 21 °C. After 120 days, the retention of GFRP and HFRP bars embedded in OPC increases by 9.0 % and 3.8 %, respectively, compared to bare FRP bars. Similarly, Fig. 8b shows that at 40 °C, their retention rises by 69.7 % and 35.3 %, while Fig. 8c indicates increases of 47.2 % and 26.4 % at 60 °C. These findings highlight the protective effect of mortar wrapping in enhancing the durability of both GFRP and HFRP bars. When comparing the mortar wrapping and carbon fiber wrapping, mortar wrapping is more effective than carbon fiber wrapping in enhancing the durability of the core GFRP bar. In conclusion, for both GFRP and HFRP bars, direct saline-alkaline solution immersion results in more severe durability degradation compared to the mortar-encased environment. In the case of bare FRP bars, where the entire bar is exposed to aggressive media, the degradation is more extensive. In contrast, mortar wrapping leads to localized, concentrated corrosion at specific points due to the pore

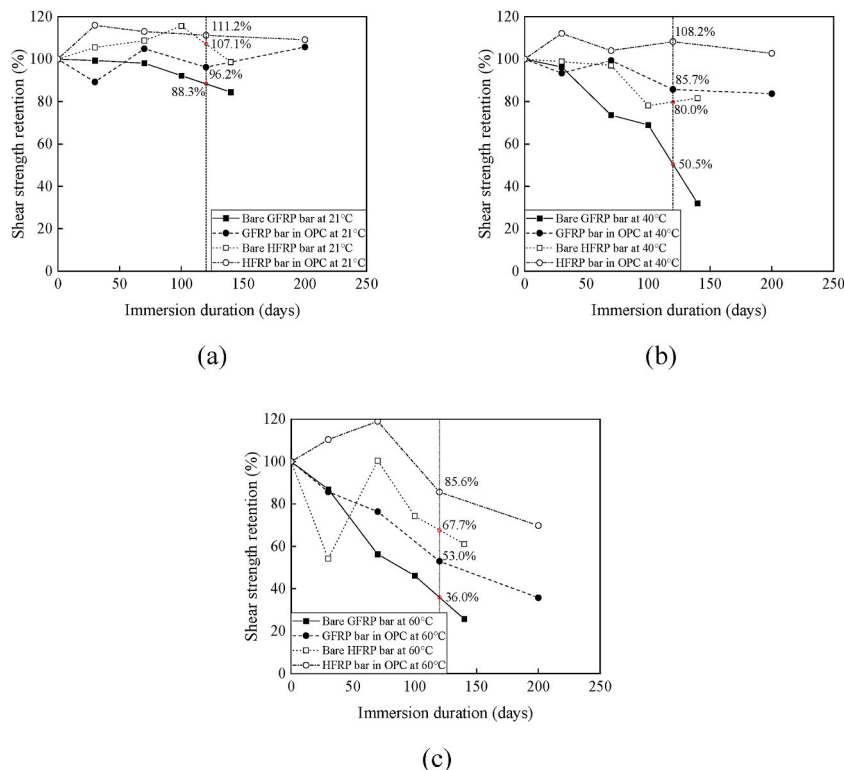


Fig. 8. Effects of mortar wrapping on the interlaminar shear strength.

solution within the concrete. Furthermore, the movement of the pore solution is restricted by the pore structure, reducing the extent of FRP bar degradation. The similar results were reported in Ref. Wu et al. [32].

3.5. Degradation mechanism

Fig. 9a illustrates the evolution of the pore structure of the mortar in seawater. In the case of the OPC mortar, the change in the total porosity of both unaged and aged OPC mortar is similar. Fig. 9b reveals that the peak of the pore size distribution for unaged OPC mortar is mainly concentrated around 100–200 nm. With increasing environmental temperatures, this peak gradually shifts to the left, indicating an increase in the number of smaller pores in OPC mortar after 200 days of immersion at 20 °C and 60 °C. This suggests that structures of OPC mortar becomes denser, thereby reducing the localized corrosion of OH^- ions on the FRP bars within the mortar. Fig. 9a presents the total porosity variation in MPC mortar. At 40 °C, the total porosity remains largely unchanged. However, at 60 °C, a notable reduction in porosity is observed compared to unaged MPC mortar. Fig. 9c indicates that after 200 days of seawater immersion at 60 °C, the peak of the pore size distribution in MPC mortar shifts below 1000 nm, indicating a more compact pore structure at this elevated temperature. In comparison with the unaged MPC mortar, at 40 °C, the number of pores smaller than 10 nm significantly decreases. This indicates an increase in gel pore morphology, which contributes to the low-alkali nature of MPC mortar.

Fig. 10 presents the SEM images of FRP bar embedded in mortar after seawater immersion. Fig. 10a shows the SEM image of an unaged GFRP bar. The interlaminar shear failure takes place within the resin, where a substantial amount remains attached to the fibers, demonstrating strong adhesion at the fiber-resin interface. Fig. 10b presents the SEM image of GFRP bars aged for 200 days in seawater at 21 °C. A significant amount of resin still adheres to the fiber surface, indicating that the 21 °C seawater immersion has little impact on the fiber-resin bond. Fig. 10c shows the SEM image of GFRP bars embedded in OPC after 200 days of exposure to 60 °C seawater. The surface fibers appear smooth, and debonding at the fiber-resin interface is evident. Elevated temperature seawater immersion causes a shift in the interlaminar shear failure mode from resin failure to fiber-resin interface failure. Fiber-resin interface debonding causes a more substantial reduction in the interlaminar shear strength of GFRP bars than resin failure. This behavior is consistent with the decline in interlaminar shear strength observed in OPC-wrapped GFRP bars under seawater exposure. When wrapped in OPC mortar, elevated temperatures accelerate the reaction of OH^- ions with the epoxy resin and the -Si-O-Si network in glass fibers, weakening the fiber-resin interface bond and ultimately reducing the overall interlaminar shear strength of the GFRP bars.

Fig. 10d shows that after 200 days in 21 °C seawater, the fiber surface of GFRP bars embedded in MPC still retains a significant amount of resin matrix. Fig. 10e shows the SEM image of GFRP bars after 200 days of immersion in 60 °C seawater, with the fiber surface still covered with a considerable amount of resin matrix. It can be observed that under MPC mortar wrapping, the environmental temperature has minimal impact on the performance of the GFRP bars. This is mainly because MPC mortar has a low pH, and as

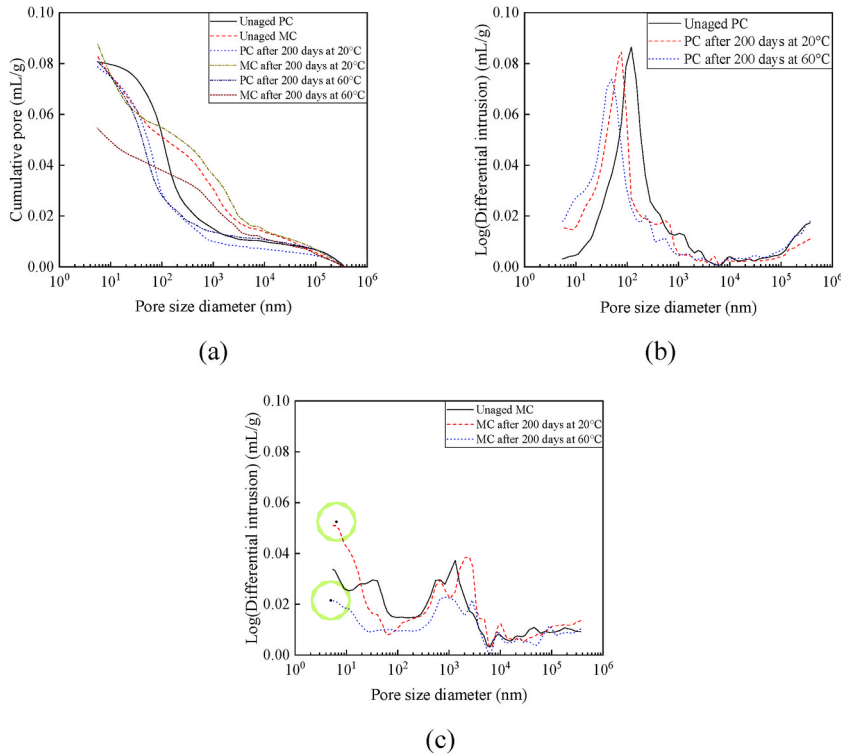


Fig. 9. Pore structure characteristics of mortar, (a) cumulative distribution, (b) differential distribution of OPC mortar, (c) differential distribution of MPC mortar.

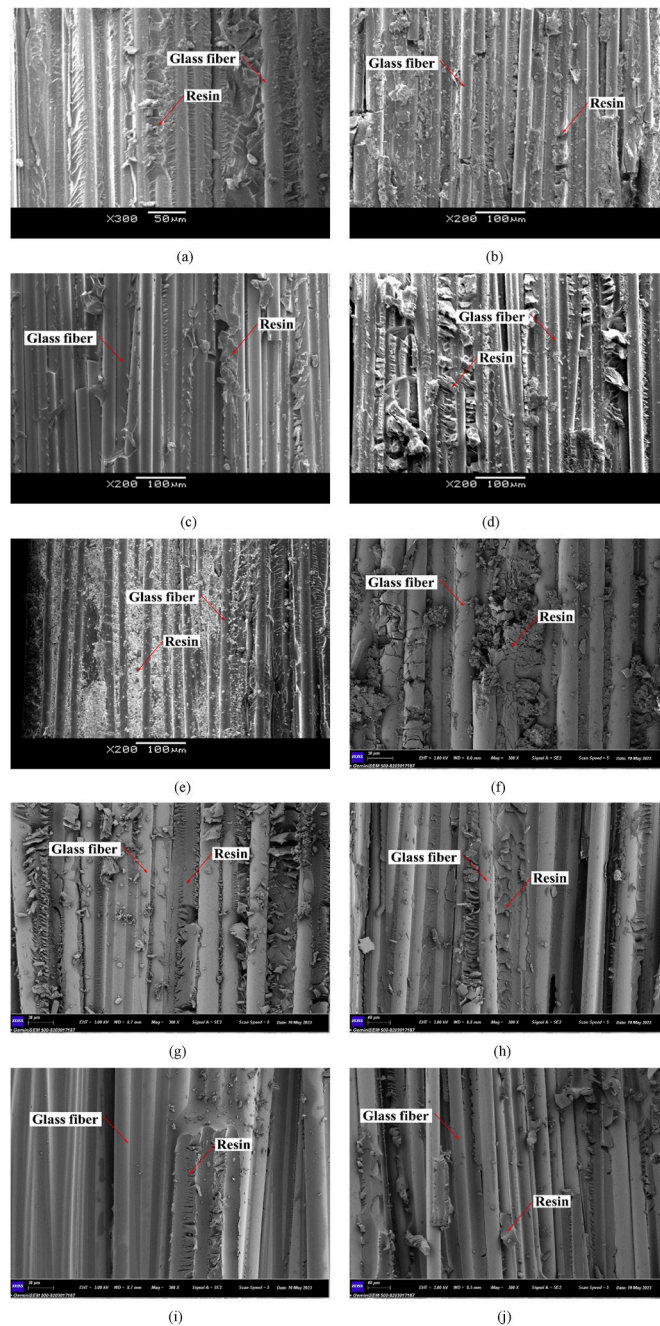


Fig. 10. SEM images of GFRP bars under various conditions, (a) control GFRP bar, (b) G-21-P-200, (c) G-60-P-200, (d) G-21-M-200, (e) G-60-M-200, (f) control HFRP bar, (g) H-21-P-200, (h) H-60-P-200, (i) H-21-M-200 and (j) H-60-M-200.

the soaking temperature increases, the gel pores (less than 10 nm) in the MPC mortar decrease, making it more difficult for OH^- ions to penetrate. Consequently, seawater immersion has minimal impact on the interlaminar shear strength.

Fig. 10f–h shows the SEM images of HFRP bars embedded in OPC mortar before aging and after 200 days of aging in seawater. Comparing Fig. 10f, g, and h, a significant amount of resin remains on the fiber surface, indicating that both 21 °C and 60 °C soaking environments have minimal effect on the fiber-resin interface bond in HFRP bars wrapped in OPC mortar. This is primarily due to the carbon fiber surface layer, which blocks the contact between OH^- ions and the core layer of glass fibers. It can be seen that under OPC mortar encapsulation, the durability of the core glass fiber bars is effectively improved by the carbon fiber wrapping around the outer layer.

Fig. 10i and j shows the SEM images of HFRP bars embedded in MPC after 200 days of aging in seawater at 21 °C and 60 °C,

respectively. Whether aged in 21 °C or 60 °C seawater for 200 days, seawater immersion does not affect the failure mode of HFRP bars, which occurs through resin matrix shear failure. This suggests that wrapping carbon fiber around the core glass fiber layer and encasing the bars in low-alkali MPC mortar effectively enhances the durability of the core glass fiber bars.

Fig. 10c and e presents SEM images of GFRP bars embedded in OPC and MPC after immersion in 60 °C seawater. The failure occurs within the resin matrix instead of at the fiber-resin interface for GFRP bars embedded in MPC. In contrast, for those embedded in OPC, the failure happens within the resin matrix itself. After 200 days, GFRP bars embedded in MPC exhibits a 201.5 % higher interlaminar shear strength than those in OPC mortar. Similarly, Fig. 10h and j shows that MPC wrapping increased HFRP bar shear strength by 28.8 %. This durability improvement of mortar wrapping is mainly attributed to the denser pore structure of OPC and MPC mortar. The reduction in gel pores (<100 nm) in MPC further lowers the pore solution pH, while the decrease in capillary pores (100–1000 nm) in OPC mortar minimizes localized erosion.

Fig. 10c and h presents SEM images of GFRP and HFRP bars embedded in OPC after 200 days of immersion in 60 °C seawater. With carbon fiber wrapped around the core glass fiber layer, the failure mode transitions from fiber-resin interface debonding to shear failure within the fiber-resin matrix, leading to a 94.9 % increase in interlaminar shear strength for HFRP bars compared to GFRP bars. Fig. 10e and j shows SEM images of GFRP and HFRP bars embedded in MPC after 200 days of exposure to 60 °C seawater. In both cases, failure occurs as shear failure within the resin matrix. After 200 days of immersion, the interlaminar shear strength of GFRP and HFRP bars retains 109.6 % and 90.9 % of their initial values, respectively. These findings indicate that MPC mortar wrapping effectively preserves both GFRP and HFRP bars with minimal degradation. Additionally, carbon fiber wrapping around the core glass fiber layer has a limited impact on enhancing the interlaminar shear strength of GFRP bars.

3.6. Prediction of the long-term interlaminar shear strength

The Arrhenius theory is commonly used to predict the long-term behavior of FRP bars. This theory suggests that as the aging temperature rises, the degradation rate of the FRP bars increases, while the underlying mechanisms driving the deterioration remain

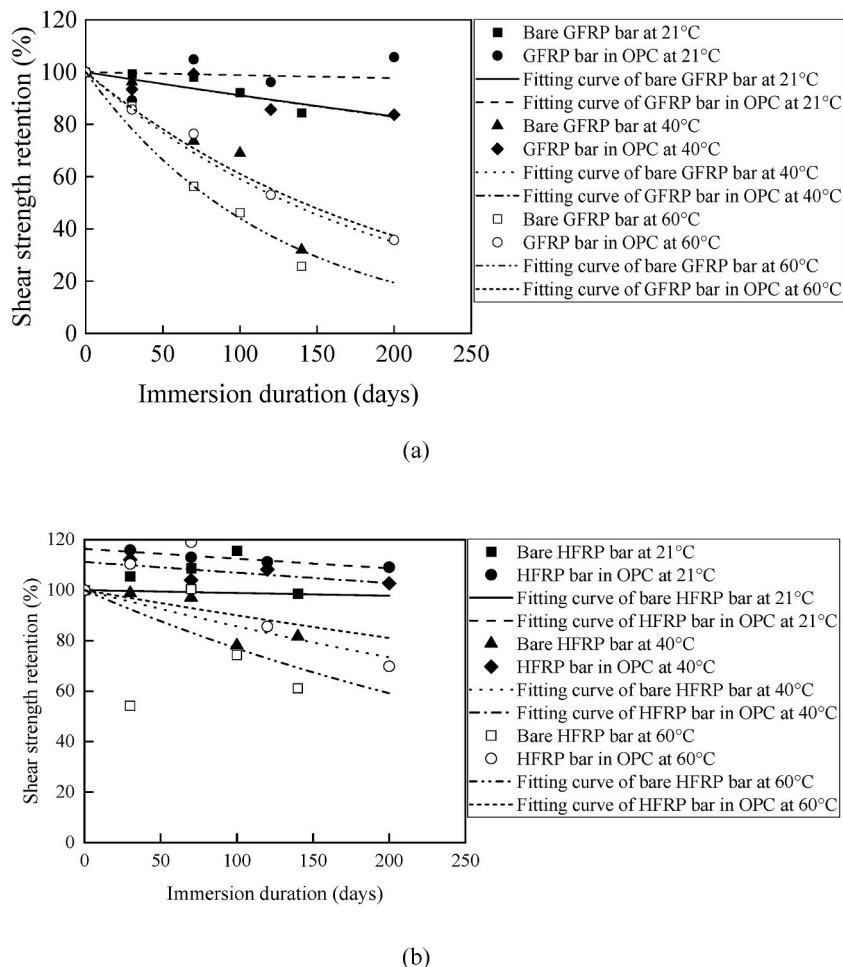


Fig. 11. Experimental results and fitting curves for GFRP (a) and HFRP (b) bars.

consistent throughout the process [32–34]. Studies have demonstrated that higher environmental temperatures lead to a decline in the interlaminar shear strength of GFRP and HFRP bars when immersed in seawater, while the factors controlling their mechanical performance stay unchanged during aging [34]. As previously discussed, seawater immersion has minimal effect on the mechanical performance of FRP bars in MPC, with the environmental temperature slightly accelerating the degradation rate of the mechanical performance. Thus, the prediction for FRP bars in MPC has not been taken into account. The Arrhenius equation is expressed as follows:

$$k = B \exp(-E_{am} / R_1 T) \quad (2)$$

where k represents the degradation rate of FRP bars when immersed in a corrosive solution (1/time). B is represents a constant related to the FRP bar material and its degradation process under a corrosive solution immersion. E_{am} is the activation energy (J/mol). T represents environmental temperature in Kelvin (K).

Taking the reciprocal of both sides of equation (2) and then taking the logarithm, equation (2) becomes:

$$\ln\left(\frac{1}{k}\right) = \frac{E_{am}}{R_1} \frac{1}{T} - \ln B \quad (3)$$

Previous studies have shown that the following equation (4) is used to describe the evolution of the mechanical properties of FRP bars over time in a corrosive environment [6,15]:

$$Y = 100e^{(-t/\tau)} \quad (4)$$

where Y represents the residual interlaminar shear strength as a percentage of the control value, and t denotes the immersion duration in the corrosive solution (days). The parameter τ is obtained through experimental data fitting. Fig. 11 displays the test results for GFRP and HFRP bars, along with the fitted curves derived from equation (4). The corresponding τ values are provided in Table 3. The findings reveal that as environmental temperature rises, shear strength retention declines. This pattern suggests that both bare FRP bars and those embedded in OPC undergo a reduction in interlaminar shear strength with increasing immersion temperature, leading to accelerated mechanical degradation.

At 21 °C, 40 °C, and 60 °C, the immersion durations required for interlaminar shear strength retention levels of 60 %, 70 %, 80 %, and 90 % were determined using equation (4) along with the values from Table 3. These results were then plotted in Fig. 12, with the vertical axis representing $\ln(1/k)$ and the horizontal axis denoting $1000/T$. By fitting the data points using equation (3), it is evident that the resulting straight lines at the three temperatures run parallel, confirming that equation (4) effectively predicts the degradation of FRP bars' mechanical properties in laboratory-accelerated tests. The E_{am}/R_1 values obtained from equation (3) fitting are provided in Table 4.

According to the Arrhenius theory, equation (2) can be used to determine the immersion times t_0 and t_1 required for the mechanical performance of FRP bars to degrade to the same level (denoted as c) at environmental temperatures T_1 and T_0 , respectively. The time-shift factor (TSF) is defined as the ratio of t_0 to t_1 , providing a measure of the acceleration effect of elevated temperatures on FRP bar degradation. The TSF calculation formula is expressed as follows [35]:

$$\text{TSF} = \frac{t_0}{t_1} = \frac{c/k_0}{c/k_1} = \frac{k_1}{k_0} = \frac{B \exp(-E_{am}/R_1 T_1)}{B \exp(-E_{am}/R_1 T_0)} = \exp\left[\frac{E_{am}}{R_1} \left(\frac{1}{T_0} - \frac{1}{T_1}\right)\right] \quad (5)$$

Field data on the interlaminar shear strength of GFRP bars in bridge concrete were reported in Refs. [36,37]. The Sierrita de la Cruz Creek Bridge in America and the Val-Alain Bridge in Canada, have mean service temperatures of 4 °C and 13.9 °C, respectively. In addition, an experiment conducted in the city of Hangzhou, with a mean temperature of 17 °C, was also considered. These three temperatures—4 °C, 13.9 °C, and 17 °C—were selected for predicting the long-term performance of FRP bars. Based on the environmental temperature used at present study and the value of E_{am}/R_1 in Table 4, the TSF for the FRP bars at 17 °C, 13.9 °C and 4.0 °C is determined by equation (5), and the TSF values are presented in Table 5.

The interlaminar shear strength over time at selected temperatures of 17 °C, 13.9 °C, and 4 °C can be determined by scaling the

Table 3
Parameters fitted by equation (4).

Specimens	Environmental temperature(°C)	τ
Bare GFRP bar	21	1080
	40	190
	60	122
Bare HFRP bar	21	8823
	40	646
	60	381
GFRP bar in OPC	21	8550
	40	1066
	60	203
HFRP bar in OPC	21	2916
	40	2550
	60	951

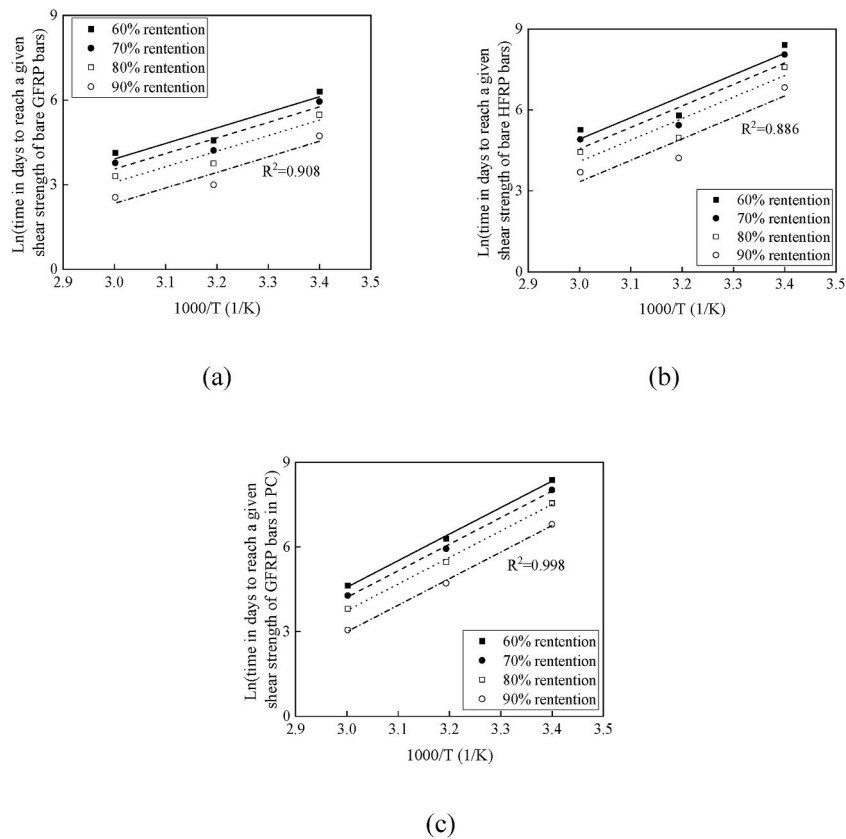


Fig. 12. The lines between $\ln(t)$ and $1000/T$ of (a) bare GFRP bar, (b) bare HFRP bar, and (c) GFRP bar embedded in OPC.

Table 4
Parameters fitted by equation (3).

Specimens	Shear strength retention (%)	E_{am}/R_1	R^2
Bare GFRP bar	60	6729	0.908
	70	6729	0.908
	80	6729	0.908
	90	6729	0.908
Bare HFRP bar	60	9627	0.886
	70	9627	0.886
	80	9627	0.886
	90	9627	0.886
GFRP bar in OPC	60	9408	0.998
	70	9408	0.998
	80	9408	0.998
	90	9408	0.998
HFRP bar in OPC	60	10148	–
	70	10148	–
	80	10148	–
	90	10148	–

Note: The mechanical performance of the HFRP bar embedded in OPC showed almost no change during the test cycle at 21 °C and 40 °C. The value of τ , calculated based on 20 °C and 40 °C, is 10148.

immersion time at 21 °C, 40 °C, and 60 °C using the corresponding TSF values. Fig. 13 presents the predicted curves of FRP bar interlaminar shear strength with time at 17 °C, 13.9 °C, and 4 °C.

Field studies on GFRP bars from the Sierrita de la Cruz Creek Bridge and the Val-Alain Bridge reveal contrasting trends in interlaminar shear strength over time. After 11 years in service at 4 °C, the GFRP bars in the Val-Alain Bridge exhibited a 14 % reduction in interlaminar shear strength [36]. In contrast, the GFRP bars in the Sierrita de la Cruz Creek Bridge, subjected to an average temperature of 13.9 °C, experienced a 4 % increase after 15 years of service [37]. Table 6 presents a comparison between the predicted and measured interlaminar shear strength of GFRP bars in both bridges. Predictions for bare GFRP bars exposed to a saline-alkaline

Table 5
Time-shift factor for FRP bars.

Type	Temperature (°C)	Time-shift factor (TSF)		
		Hangzhou (17 °C)	Sierrita de la Cruz Creek Bridge (13.9 °C)	Val-Alain Bridge (4.0 °C)
Bare GFRP bar	21	1.4	1.8	4.1
	40	5.5	7.1	16.3
	60	20	25.6	59.2
Bare HFRP bar	21	1.6	2.2	7.4
	40	11.4	16.4	54.2
	60	72.4	103.6	343.4
GFRP bar embedded in OPC	21	1.6	2.2	7.1
	40	10.8	15.4	49.5
	60	65.7	93.2	300.7
HFRP bar embedded in OPC	21	1.6	2.3	8.3
	40	13.1	19	67.3
	60	91.3	133.2	471

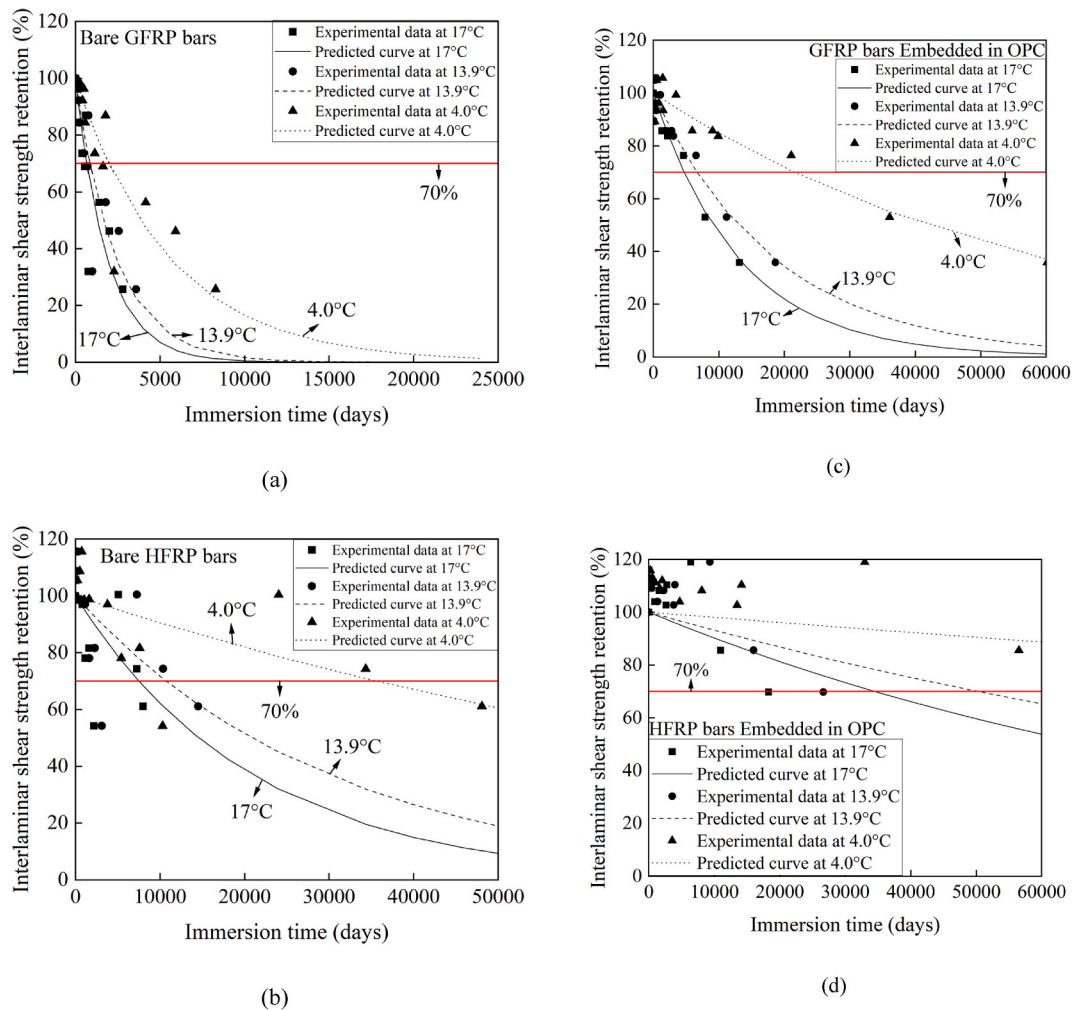


Fig. 13. Predicted curves for (a) bare GFRP bars and (b) bare HFRP bars, (c) GFRP bars embedded in OPC, and (d) HFRP bars embedded in MPC.

solution indicate a significant reduction of 51 % after 11 years at 4 °C and 90 % after 15 years at 13.9 °C. However, these predictions tend to overestimate the actual degradation observed in field results. Conversely, when embedded in OPC mortar, the interlaminar shear strength of GFRP bars is expected to decline by just 6 % over 11 years at 4 °C and by 25 % after 15 years at 13.9 °C. Notably, in the Sierrita de la Cruz Creek Bridge, the interlaminar shear strength of the GFRP bars actually increased by 4 %, suggesting that post-curing effects outweigh the degradation caused by OH⁻ ions.

Table 6

Comparison between the predicted and measured interlaminar shear strength of GFRP bars in both bridges.

Bridge	Annual mean temperature (°C)	Service time (years)	Shear strength retention (%)					
			Field results		Predicted results			
			Value	CV	Bare GFRP bar	Bare HFRP bar	GFRP bar embedded in OPC	HFRP bar embedded in OPC
Val-Alain Bridge in Canada	4.0 °C	11	86 %	0.02	49 %	96 %	94 %	99 %
Sierrita de la Cruz Creek Bridge in USA	13.9 °C	15	104 %	–	10 %	83 %	75 %	96 %

The ACI 440.1R-2015 code specifies that GFRP bars embedded in concrete with high humidity (90 % RH) should retain at least 70 % of their shear strength. The time required to reach this threshold is determined by where the 70 % retention line intersects with the predicted degradation curves in Fig. 13, with results provided in Table 7. For bare GFRP bars, this level is reached in 1.8–5.4 years, while for bare HFRP bars, the time ranges from 20.6 to 97.5 years when exposed to alkaline-seawater environments at temperatures between 4 °C and 17 °C. This demonstrates that the carbon fiber coating on GFRP bars significantly enhances their long-term performance. Furthermore, the presence of OPC mortar around GFRP and HFRP bars extends their shear strength retention to 12.9–59.0 years for GFRP bars and 94.5–487.6 years for HFRP bars, underscoring the critical role of OPC mortar in improving the durability of FRP bars in seawater conditions.

Comparisons indicate that both the carbon fiber coating on GFRP bars and the OPC mortar wrapping significantly boost the durability of the bars in seawater conditions. However, the replacement of OPC mortar with MPC mortar, which reduces the pH value in mortar, results in better durability for FRP bars wrapped in MPC mortar compared to those wrapped in OPC mortar. The long-term predictions based on the experimental results of bare FRP bars in saline-alkaline solutions are conservative, while those for FRP bars embedded in OPC mortar in seawater underestimate their degradation. Future studies should exclude the effects of elevated temperature on the wrapping mortar.

The present study focuses on the durability of FRP bars embedded in mortar, whereas the durability of specimens with FRP bars wrapped around mortar should be addressed in future research.

4. Conclusion

This study examines the durability of GFRP and HFRP bars embedded in OPC and MPC mortars under simulated seawater conditions. It highlights the protective effects of carbon fiber coatings and mortar encasement in enhancing the durability of GFRP bars. Using Arrhenius theory, long-term durability predictions are compared with real-world environmental performance data. The key conclusions drawn from this research are as follows.

- 1) Reducing the alkalinity of the internal service environment and utilizing protective carbon-fiber coatings to GFRP bars significantly enhance the saline-alkaline resistance of GFRP bars.
- 2) Both GFRP and HFRP bars embedded in either OPC or MPC experienced failure due to the propagation of the main crack along the neutral axis. With prolonged immersion and increasing temperature, failure shifts from resin matrix shear failure to fiber-resin interface debonding.
- 3) GFRP bars in OPC mortar experience greater interlaminar shear strength loss than those in MPC. After 200 days, MPC-embedded bars exhibit -5.7 %, 17.2 %, and 201.5 % higher strength retention than OPC-embedded bars at 21 °C, 40 °C, and 60 °C, respectively.
- 4) Carbon fiber coatings significantly enhance the durability of GFRP bars in seawater-exposed mortar. HFRP bars retain 3.3 % more strength than GFRP bars at 21 °C, 22.7 % more at 40 °C, and 94.9 % more at 60 °C after 200 days.
- 5) OPC mortar improves interlaminar shear strength retention compared to direct exposure to saline-alkaline solutions. After 120 days at 60 °C, strength retention increases by 47.2 % for GFRP bars and 26.4 % for HFRP bars.
- 6) The interlaminar shear strength of the HFRP bar embedded in MPC mortar remains stable after 200 days of seawater immersion at 20 °C and 40 °C.

Table 7

Predicted results of FRP bars at 17 °C, 13.9 °C and 4 °C.

Annual mean temperature (°C)	Time in years to reach 70 % shear strength retention			
	Bare GFRP bar	Bare HFRP bar	GFRP bar embedded in OPC	HFRP bar embedded in OPC
17 °C	1.8	20.6	12.9	94.5
13.9 °C	2.5	29.4	18.3	137.8
4 °C	5.4	97.5	59.0	487.6

- 7) The time required to reach 70 % interlaminar shear strength retention at 13.9 °C is estimated at 2.5 years for bare GFRP bars, 29.4 years for bare HFRP bars, 18.3 years for GFRP bars embedded in OPC mortar, and 137.8 years for HFRP bars embedded in OPC mortar.

CRedit authorship contribution statement

Yunfeng Pan: Writing – original draft, Methodology, Investigation, Funding acquisition. **Junmin Wang:** Writing – original draft, Investigation, Formal analysis. **Angelo Paulo Mogul:** Writing – review & editing, Writing – original draft. **Jun He:** Writing – review & editing, Funding acquisition. **Gaojie Xu:** Investigation.

Data availability statement

All data are available in the main text.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The authors would like to acknowledge the financial support from the National Natural Science Foundation of China [grant numbers 52278280], and Fundamental Research Funds for the Central Universities (grant numbers 300102283502).

Data availability

Data will be made available on request.

References

- [1] B. Benmokrane, A.H. Ali, Durability and long-term performance of fiber-reinforced polymer as a new civil engineering material, Springer International Publishing, Cham, 2018, pp. 49–59.
- [2] B. Wang, C. Cui, C. Xu, K. Meng, J. Li, L. Xu, A novel analytical solution for horizontal vibration of partially embedded offshore piles considering the distribution effect of wave loads, Ocean. Eng. 307 (2024) 118179, <https://doi.org/10.1016/j.oceaneng.2024.118179>.
- [3] C.A. Apostolopoulos, The influence of corrosion and cross-section diameter on the mechanical properties of B500c steel, J. Mater. Eng. Perform. 18 (2) (2009) 190–195, <https://doi.org/10.1007/s11665-008-9281-x>.
- [4] C. Lu, M. Ni, T. Chu, L. He, Comparative investigation on tensile performance of FRP bars after exposure to water, seawater, and alkaline solutions, J. Mater. Civ. Eng. 32 (7) (2020) 04020170, [https://doi.org/10.1061/\(ASCE\)MT.1943-5533.0003243](https://doi.org/10.1061/(ASCE)MT.1943-5533.0003243).
- [5] W. Wu, X. He, W. Yang, M.S. Alam, B. Wei, J. He, Degradation factors and microstructure degradation characteristics of B/GFRP bars in harsh environment: a review, Constr. Build. Mater. 366 (2023) 130246, <https://doi.org/10.1016/j.conbuildmat.2022.130246>.
- [6] Y. Pan, D. Yan, Study on the durability of GFRP bars and carbon/glass hybrid fiber reinforced polymer (HFRP) bars aged in alkaline solution, Compos. Struct. 261 (2021) 113285, <https://doi.org/10.1016/j.compstruct.2020.113285>.
- [7] P. Wang, L.-y.-w. Ke, H.-l. Wu, C.K.Y. Leung, Lowering exposure pH for durability enhancement of glass fiber reinforcement polymer (GFRP) rebars, Constr. Build. Mater. 354 (2022) 129131, <https://doi.org/10.1016/j.conbuildmat.2022.129131>.
- [8] Z. Jiang, B. Zhang, C. Zhao, J. Zhao, Z. Lu, J. Fang, J. Xie, Long-term shear performance of prestressed GFRP bars in accelerated seawater aging: role of alkalinity reduction in concrete matrix, Constr. Build. Mater. 463 (2025) 140039, <https://doi.org/10.1016/j.conbuildmat.2025.140039>.
- [9] G. Feng, D. Zhu, S. Guo, M.Z. Rahman, Z. Jin, C. Shi, A review on mechanical properties and deterioration mechanisms of FRP bars under severe environmental and loading conditions, Cement Concr. Compos. 134 (2022) 104758, <https://doi.org/10.1016/j.cemconcomp.2022.104758>.
- [10] L. Yuan, H. Wang, T. He, S. Gao, Review on the effect collision between hazardous metal ions and geopolymer as adsorbents or in situ stabilization/solidification, Appl. Clay Sci. 249 (2024) 107258, <https://doi.org/10.1016/j.clay.2024.107258>.
- [11] C. Lu, Z. Qi, Y. Zheng, G. Xuan, Y. Yan, Long-term tensile performance of GFRP bars in loaded concrete and aggressive solutions, J. Build. Eng. 64 (2023) 105587, <https://doi.org/10.1016/j.job.2022.105587>.
- [12] S.-Z. Feng, J.-J. Zeng, B. Zhao, Z.-H. Hao, Y. Zhuge, Q.-M. Zhong, Z.-W. Zhang, Accelerated aging tests of large-diameter GFRP bars in alkaline environment, Composites Part C: Open Access 14 (2024) 100486, <https://doi.org/10.1016/j.jcom.2024.100486>.
- [13] B. Benmokrane, A.H. Ali, H.M. Mohamed, A. ElSafty, A. Manalo, Laboratory assessment and durability performance of vinyl-ester, polyester, and epoxy glass-FRP bars for concrete structures, Compos Part B 114 (2017) 163–174, <https://doi.org/10.1016/j.compositesb.2017.02.002>.
- [14] G. Mai, Z. Xiong, H. Zhu, L. Zhou, H. Zhou, L. Li, Durability of GFRP bars embedded in seawater sea sand concrete in marine environments, Constr. Build. Mater. 458 (2025) 139488, <https://doi.org/10.1016/j.conbuildmat.2024.139488>.
- [15] Z. Wang, X.-L. Zhao, G. Xian, G. Wu, R.S. Raman, S. Al-Saadi, Durability study on interlaminar shear behaviour of basalt-, glass-and carbon-fibre reinforced polymer (B/G/CFRP) bars in seawater sea sand concrete environment, Constr. Build. Mater. 156 (2017) 985–1004, <https://doi.org/10.1016/j.conbuildmat.2017.09.045>.
- [16] G. Xian, P. Zhou, C. Li, S. Dong, H. Du, J. Tian, R. Guo, Z. Peng, Z. Zhang, T. He, Mechanical properties evaluation of glass fiber reinforced thermoplastic composite plate under combined bending loading and water immersion, Constr. Build. Mater. 440 (2024) 137470, <https://doi.org/10.1016/j.conbuildmat.2024.137470>.
- [17] C. Li, X. Yin, Y. Wang, L. Zhang, Z. Zhang, Y. Liu, G. Xian, Mechanical property evolution and service life prediction of pultruded carbon/glass hybrid rod exposed in harsh oil-well condition, Compos. Struct. 246 (2020) 112418, <https://doi.org/10.1016/j.compstruct.2020.112418>.
- [18] Y. Yu, S. Liu, Y. Pan, X. Miu, J. Liu, Durability of glass fiber-reinforced polymer bars in water and simulated concrete pore solution, Constr. Build. Mater. 299 (2021) 123995, <https://doi.org/10.1016/j.conbuildmat.2021.123995>.
- [19] C. Xiong, Y. Huang, Z. Jin, P. Wang, Y. Hu, Influence of salt concentration and pH value on the degradation of epoxy resin used in FRP bars under simulated marine conditions, J. Build. Eng. 98 (2024) 111405, <https://doi.org/10.1016/j.job.2024.111405>.

- [20] A. Manalo, G. Maranan, B. Benmokrane, P. Cousin, O. Alajarmeh, W. Ferdous, R. Liang, G. Hota, Comparative durability of GFRP composite reinforcing bars in concrete and in simulated concrete environments, *Cement Concr. Compos.* 109 (2020) 103564, <https://doi.org/10.1016/j.cemconcomp.2020.103564>.
- [21] M. Robert, P. Cousin, B. Benmokrane, Durability of GFRP reinforcing bars embedded in moist concrete, *J Compos Constr* 13 (2) (2009) 66–73, [https://doi.org/10.1061/\(ASCE\)1090-0268\(2009\)13:266](https://doi.org/10.1061/(ASCE)1090-0268(2009)13:266).
- [22] Y. Swolfs, L. Gorbatiikh, I. Verpoest, Fibre hybridisation in polymer composites: a review, *Compos. Appl. Sci. Manuf.* 67 (2014) 181–200, <https://doi.org/10.1016/j.compositesa.2014.08.027>.
- [23] G. Xu, Y. Yu, Y. Pan, B. Li, Durability of bond between carbon/glass hybrid fiber-reinforced polymer (HFRP) bar and concrete in water, *Adv. Struct. Eng.* 26 (9) (2023) 1752–1764, <https://doi.org/10.1177/13694332231175391>.
- [24] X. Guo, Z. Jin, C. Xiong, T. Sun, N. Li, Y. Yu, X. Zhang, Deterioration of mechanical properties of basalt/carbon hybrid FRP bars in SWSC under seawater corrosive environment, *Constr. Build. Mater.* 317 (2022) 125979, <https://doi.org/10.1016/j.conbuildmat.2021.125979>.
- [25] Y. Pan, J. Shi, Y. Yu, G.J. Xian, Effects of marine environment on water diffusion and interlaminar shear strength of carbon-glass hybrid-fiber reinforced polymer bars, *Ind. Constr.* 53 (4) (2023) 161–172, <https://doi.org/10.13204/j.gyjzG21062705>.
- [26] Z. Lu, Y. Li, J. Xie, Durability of BFRP bars wrapped in seawater sea sand concrete, *Compos. Struct.* 255 (2021) 112935, <https://doi.org/10.1016/j.compstruct.2020.112935>.
- [27] Y. Pan, G. Xian, H. Li, Numerical modeling of moisture diffusion in an unidirectional fiber-reinforced polymer composite, *Polym. Compos.* 40 (1) (2019) 401–413, <https://doi.org/10.1002/pc.24664>.
- [28] Y. Chen, J.F. Davalos, I. Ray, H.-Y. Kim, Accelerated aging tests for evaluations of durability performance of FRP reinforcing bars for concrete structures, *Compos. Struct.* 78 (1) (2007) 101–111, <https://doi.org/10.1016/j.compstruct.2005.08.015>.
- [29] E. Berodier, J. Bizzozero, A. C. A. Muller, Mercury intrusion porosimetry, in: K. Scrivener, R. Snellings, B. Lothenbach (Eds.), *A practical guide to microstructural analysis of cementitious materials*, CRC Press Taylor & Francis Group, Boca Raton, 2016, pp. 419–444.
- [30] W. Sun, Y. Zheng, L. Zhou, J. Song, Y. Bai, A study of the bond behavior of FRP bars in MPC seawater concrete, *Adv. Struct. Eng.* 24 (6) (2021) 1110–1123, <https://doi.org/10.1177/1369433220956816>.
- [31] Z. Wang, X.-L. Zhao, G. Xian, G. Wu, R.K.S. Raman, S. Al-Saadi, Effect of sustained load and seawater and sea sand concrete environment on durability of basalt-and glass-fibre reinforced polymer (B/GFRP) bars, *Corros. Sci.* 138 (2018) 200–218, <https://doi.org/10.1016/j.corsci.2018.04.002>.
- [32] Z. Dong, G. Wu, X.-L. Zhao, Z.-K. Wang, A refined prediction method for the long-term performance of BFRP bars serviced in field environments, *Constr. Build. Mater.* 155 (Supplement C) (2017) 1072–1080, <https://doi.org/10.1016/j.conbuildmat.2017.07.154>.
- [33] J.F. Davalos, Y. Chen, I. Ray, Long-term durability prediction models for GFRP bars in concrete environment, *J. Compos. Mater.* 46 (16) (2012) 1899–1914, <https://doi.org/10.1177/0021998311427777>.
- [34] G. Wu, Z.-Q. Dong, X. Wang, Y. Zhu, Z.-S. Wu, Prediction of long-term performance and durability of BFRP bars under the combined effect of sustained load and corrosive solutions, *J. Compos Constr* 19 (3) (2015) 04014058, [https://doi.org/10.1061/\(ASCE\)CC.1943-5614.0000517](https://doi.org/10.1061/(ASCE)CC.1943-5614.0000517).
- [35] Z. Dong, G. Wu, B. Xu, X. Wang, L. Taerwe, Bond durability of BFRP bars embedded in concrete under seawater conditions and the long-term bond strength prediction, *Mater Design* 92 (2016) 552–562, <https://doi.org/10.1016/j.matdes.2015.12.066>.
- [36] B. Benmokrane, C. Nazair, M.-A. Loranger, A. Manalo, Field durability study of vinyl-ester based GFRP rebars in concrete bridge barriers, *J. Bridge Eng.* 23 (12) (2018) 04018094, [https://doi.org/10.1061/\(ASCE\)BE.1943-5592.0001315](https://doi.org/10.1061/(ASCE)BE.1943-5592.0001315).
- [37] O. Gooranorimi, A. Nanni, GFRP reinforcement in concrete after 15 years of service, *J Compos Constr* 21 (5) (2017) 04017024, [https://doi.org/10.1061/\(ASCE\)CC.1943-5614.0000806](https://doi.org/10.1061/(ASCE)CC.1943-5614.0000806).